

13

Potential Roadblocks while building India's fastest PC.

Even sports cars hit a few roadblocks now and then.

Siddhant Sharma

Bringing all parts required and putting them together in a build of this magnitude is not as simple as saying: "Yay! Let's build the fastest PC we can." It's always easier said than done because there are too many factors that are beyond your control.

Just sourcing the components was a pain, some of which took over six months to acquire. Factoring in the limited availability of the niche components we require, actually having them arrive in our labs at the same time was a herculean feat.

Our product coordinator was run off her feet and Big Daddy developed a few grey hairs in the process. In the end though, it was all worth it.

1 Money

The most important thing that you need to build an Ultimate PC like this is, well,... *Drumroll*.... Money, and lots of it. You can't build a PC like this even if you have a few pots of gold with you because what is required is at least a medium sized treasure chest. Money makes the world go round and when we saw the total amount of money the parts in this build cost, half of us nearly fainted. They said "Money can't buy happiness", but once we built the Ultimate PC, we figured they were all wrong. The Price vs. Performance ratio of this build is skewed beyond imagination considering you can easily get away with building six of our "ultra-high end" rigs (we should really change that name now) from our killer rigs section.

2 Pairing two PSUs

Ed: "Hey Joe, we don't think a single 1200 Watts 80-plus Platinum certified PSU would be enough, we would at least need two of those suckers."

Joe gasps in horror, drops everything in his hands and runs like the wind.

Ed: "Well that's another intern we lost this week because of this darned Ultimate PC. I won't be able to find more minions to help build this beast of a machine in time because everyone is afraid it'll blow up."

The above is an excerpt from Chapter 10: "Building the Ultimate PC" of our imaginary book titled: "Life at the Digit Test Centre". This kind of situation arises when you're putting four GTX 780 Ti's in Quad SLI on an already power hungry machine. Unless you have an easy solution to this problem in the form of two OC Link-enabled Antec HCP PSUs, you'll have to make do with building your own 24-pin adapter, which is not a task for the meek. Especially when you have ₹16,00,000 depending on your soldering skills.

3 Space constraints:

Realty prices are skyrocketing and the desire to satisfy your need for more space is a tough pill to swallow. Similarly, when you have a boatload of components to build the Ultimate PC, you'll run out of space for your components. The problem is not only with fitting multiple GPUs in the system, it's also for finding the right parts that don't encroach on more space than they need and provide more clearance. Air coolers (big ones) have been known to do that, especially making RAM installation near the CPU area almost impossible or obstructing the side-panel of a case with their height. Another factor that needs addressing is liquid cooling, will there be enough space to mount the kind of radiator you're looking at? Radiators come in plenty of different sizes and you have to be sure which one will fit and where.

4 SATA 6Gb/s limitations on the X79 chipset:

Though Intel's X79 chipset is currently the most powerful consumer chipset on the market, it has a few silly shortcomings mostly in the form of availability of Intel's native SATA 6Gb/s ports. It houses I/O functionality in the form of only two native Intel SATA 6Gb/s ports and if a user wants more such ports, he's forced to use the ASMedia based SATA 3 ports. Intel's controller has direct access to the chipset and is a faster and more stable option as far as transfer rates go, which isn't the case with third-party ASMedia controllers. This is why we used the two Intel SATA 6 Gb/s ports for RAID 0 on our SanDisk Extreme II and the other two Intel SATA 3 Gb/s ports for RAID 0 on the remaining two SanDisk SSDs.

5 Bottleneck at the CPU end in Quad SLI

When we said earlier that the Intel Core i7 4960X Extreme Edition can address a maximum of 40 PCI Express 3.0 lanes, we were being less than honest. The reason for this is that since we wanted to use Quad-SLI, the number of lanes per card would decrease and the CPU could only address the four 780 Ti's in a x16/x8/x8/x8 configuration which seriously begs the question as to why Intel didn't address this with their X79 chipset. Hopefully, they'll improve with their upcoming X99 platform.

6 Where's the good stuff?:

Even though we tried to avoid talking about this, we have to because when you want to build the fastest PC, you'll want to get your hands on some rare parts. They're easily available overseas, but are very hard to find here.